

GWP-3

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August 2025

1 Disclaimer on Use of Artificial Intelligence

This project utilised artificial intelligence (AI) in the following ways:

- To enhance understanding of the theoretical framework underpinning portfolio selection algorithms.
- To facilitate the identification and retrieval of relevant references and bibliography entries, ensuring a robust literature foundation.
- To optimise Python code for data collection, processing, and algorithm implementation, improving efficiency and accuracy.
- To assist in reviewing and refining the writing style of this report, ensuring clarity, consistency, and adherence to academic standards.

2 Pseudocode for Sequential Portfolio Selection

This section presents the pseudocode for the sequential portfolio selection problem as outlined in Model 1 of [1]. The model formulates portfolio selection as a sequential decision-making process within the multi-armed bandit framework, addressing the challenge of allocating wealth across a basket of assets over a series of trials. The process involves receiving historical returns, filtering a subset of assets, iteratively selecting portfolio weights, observing asset returns, and calculating portfolio rewards. The pseudocode below elaborates these steps, providing a clear algorithmic structure for implementation in subsequent sections.

3 Data Collection and Processing

Daily closing prices for 28 of the 30 specified S&P 500 stocks were collected for September to October 2008, using Yahoo Finance via Python's `yfinance` library. The stocks include 14 financial institutions (JPM, WFC, BAC, C, GS, USB, MS, KEY, PNC, COF, AXP, PRU, SCHW, TFC) and 14 non-financial companies (KR, PFE, XOM, WMT, DAL, CSCO, EQIX, DUK, NFLX, GE, APA, F, REGN, CMS). Data for SIVB and PEAK

Algorithm 1 Sequential Portfolio Selection Algorithm (Model 1)

- 1: **Input:** Number of historical trials δ , number of investment trials N , set of assets in the financial market
 - 2: **Output:** Sequence of portfolio weights ω_t and rewards $\omega_t^\top R_t$ for $t = 1, \dots, N$
 - 3: **Initialisation:**
 - 4: Receive historical returns $H_{i,t}$ for each asset i in the market for trials $t = 1, \dots, \delta$
 - 5: Filter the market to select a basket of K assets using a graph-theoretic approach (see Section 4 for details)
 - 6: **Main Loop:**
 - 7: **for** $t = 1, \dots, N$ **do**
 - 8: Choose portfolio weights $\omega_t = (\omega_{1,t}, \dots, \omega_{K,t})^\top$, where $\omega_t \in W = \{u \in \mathbb{R}_+^K : u^\top \mathbf{1} = 1\}$
 - 9: Observe asset returns $R_t = (R_{1,t}, \dots, R_{K,t})^\top$, where $R_{i,t} = \log(P_{i,t+1}/P_{i,t})$
 - 10: Compute portfolio reward $\omega_t^\top R_t$
 - 11: Update accumulated knowledge with observed returns R_t for use in subsequent trials
 - 12: **end for**
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were unavailable due to ticker changes or delisting. The prices were structured into a pandas DataFrame, and daily log returns were computed as $R_{i,t} = \log(P_{i,t+1}/P_{i,t})$.

The resulting DataFrame contains 42 trading days (2 September 2008 to 30 October 2008) and 28 stocks, reflecting high volatility due to the 2008 financial crisis. A time series plot of returns for JPM, BAC, NFLX, WMT, and GE, shown in Figure 1, illustrates significant fluctuations, particularly in financial stocks like BAC and JPM. The most volatile days were 29 September 2008 (sum of absolute returns: 0.6451) and 15 October 2008 (0.5753), with BAC and C showing the highest stock volatility (standard deviations: 0.0714 and 0.0700, respectively).

4 Correlation Matrix and Stock Sorting

A 28×28 correlation matrix was computed for the daily log returns of the 28 stocks over 42 trading days, using the formula from [1]: $\rho_{i,j} = \frac{\langle H_i H_j \rangle - \langle H_i \rangle \langle H_j \rangle}{\sqrt{(\langle H_i^2 \rangle - \langle H_i \rangle^2)(\langle H_j^2 \rangle - \langle H_j \rangle^2)}}$, where $\langle \cdot \rangle$ denotes the historical mean. The matrix was validated against Python's pandas correlation function, showing a maximum difference of 0.0000. The mean correlation (excluding diagonal) was 0.5986, with a standard deviation of 0.1376. The highest correlation was between XOM and APA (0.9157), and the lowest was between KEY and NFLX (0.1815).

Stocks were sorted using hierarchical clustering based on the correlation matrix, grouping those with similar correlation patterns, as inspired by [2]. This approach supports the paper's graph-theoretic methodology for selecting peripheral assets in the minimum spanning tree to reduce systemic risk [3, 4]. The sorted order (e.g., DAL, NFLX, PRU, ..., XOM, APA) revealed distinct clusters, with financial stocks (e.g., BAC, C, JPM) showing stronger correlations than non-financial stocks (e.g., NFLX, DUK). A heatmap of the sorted matrix, shown in Figure 2, and a dendrogram illustrating the clustering hierarchy, shown in Figure 3.

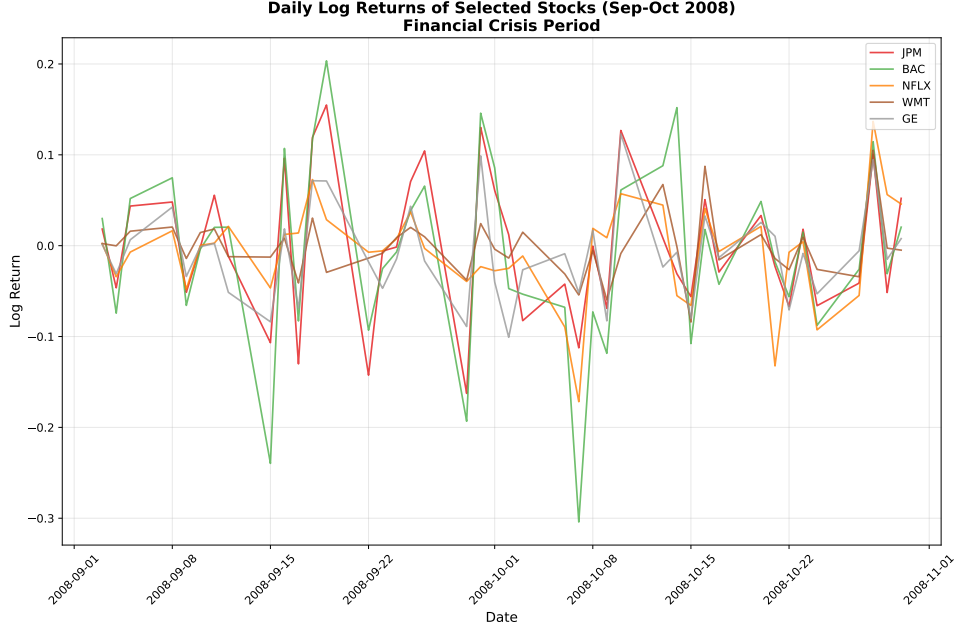


Figure 1: Time series plot of returns for JPM, BAC, NFLX, WMT, and GE, showing significant fluctuations over the 2008 financial crisis.

5 UCB1 Algorithm Implementation

The Upper Confidence Bound (UCB1) algorithm, as described in [1] (Section 2.3, Equation 2.1), was implemented to select portfolios from 28 S&P 500 stocks over 42 trading days in September to October 2008. The algorithm balances exploration and exploitation by selecting assets based on their empirical mean returns plus a confidence bound, defined as $UCB_i = \mu_i + \sqrt{\frac{2 \ln t}{n_i}}$, where μ_i is the mean return of asset i , n_i is the number of times asset i is selected, and t is the current trial.

Algorithm 2 UCB1 Algorithm for Portfolio Selection

- 1: **Input:** Trials N , assets K , returns $R_{i,t}$ for $i = 1, \dots, K$, $t = 1, \dots, N$
 - 2: **Output:** Portfolio weights ω_t , wealth S_N
 - 3: Initialise $n_i = 0$, $\mu_i = 0$ for each asset i , wealth $S_0 = 1$
 - 4: **for** $t = 1$ to N **do**
 - 5: **if** any $n_i = 0$ **then**
 - 6: Select asset i_t with $n_i = 0$, set $\omega_t = e_{i_t}$
 - 7: **else**
 - 8: Compute $UCB_i = \mu_i + \sqrt{\frac{2 \ln t}{n_i}}$ for each i
 - 9: Select $i_t = \arg \max_i UCB_i$, set $\omega_t = e_{i_t}$
 - 10: **end if**
 - 11: Observe return $R_{i_t,t}$, update $n_{i_t} \leftarrow n_{i_t} + 1$
 - 12: Update $\mu_{i_t} \leftarrow \frac{(n_{i_t}-1)\mu_{i_t} + R_{i_t,t}}{n_{i_t}}$
 - 13: Update wealth $S_t = S_{t-1} \cdot (1 + \omega_t^\top R_t)$
 - 14: **end for**
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The algorithm was applied to the daily log returns from Section 3, with an initial wealth of 1. The final cumulative wealth was 0.8126, outperforming the equal-weighted

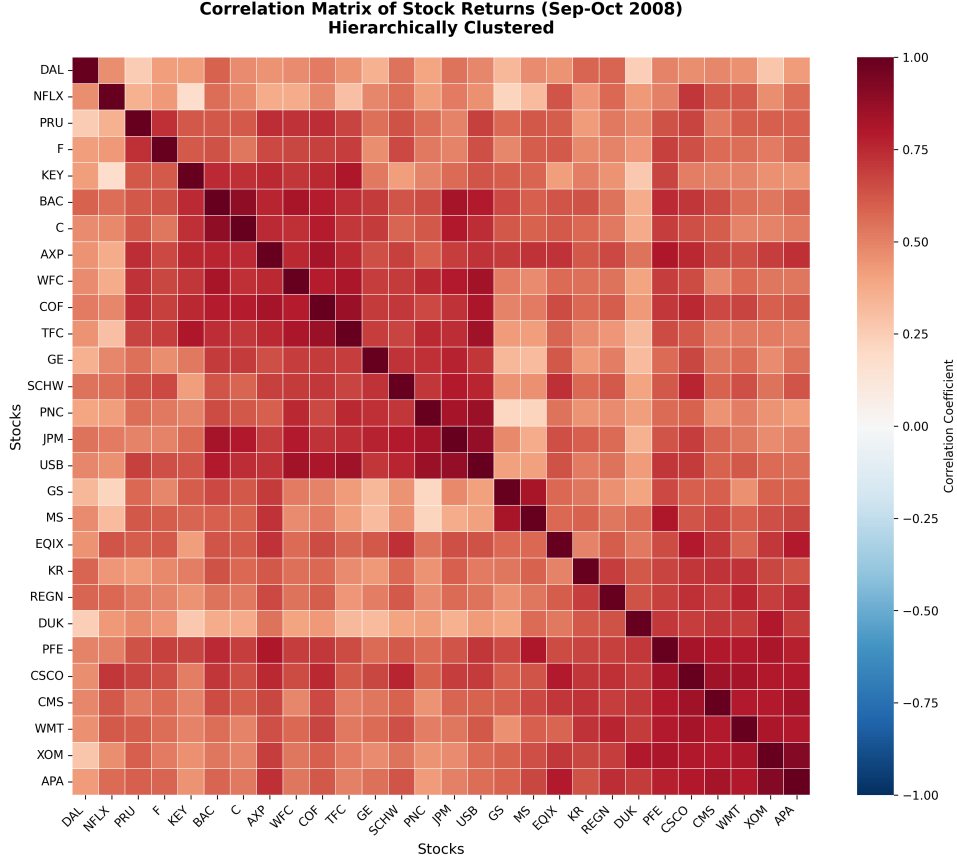


Figure 2: Heatmap of the 28×28 correlation matrix for the 28 S&P 500 stocks, sorted by hierarchical clustering.

portfolio’s wealth of 0.7305. The algorithm initially explored all 28 stocks (e.g., JPM, WFC, BAC for the first three days) before converging to assets with higher UCB values (e.g., USB, MS in later trials). A plot comparing the UCB1 and equal-weighted portfolio wealth, shown in Figure 4. The plot illustrates that the UCB1 portfolio generally maintained higher wealth, despite volatility, reflecting its ability to adaptively select assets with higher expected returns during the 2008 crisis.

6 Epsilon-Greedy Algorithm Implementation

The epsilon-greedy algorithm, as referenced in [1] (Section 3.2), was implemented to select portfolios from 28 S&P 500 stocks over 42 trading days in September to October 2008. The algorithm selects the asset with the highest empirical mean return with probability $1 - \epsilon$ and a random asset with probability ϵ , set to 0.1, balancing exploration and exploitation.

The algorithm was applied to the daily log returns from Section 3, with an initial wealth of \$1. The final cumulative wealth was \$1.0473, significantly outperforming the equal-weighted portfolio’s wealth of \$0.7305. The algorithm initially selected stocks such as JPM and WFC, later shifting to APA, BAC, and F, reflecting adaptive exploitation of high-return assets. A plot comparing the epsilon-greedy and equal-weighted portfolio wealth, shown in Figure 5. The plot highlights the epsilon-greedy portfolio’s superior performance, with notable growth despite market volatility during the 2008 crisis.

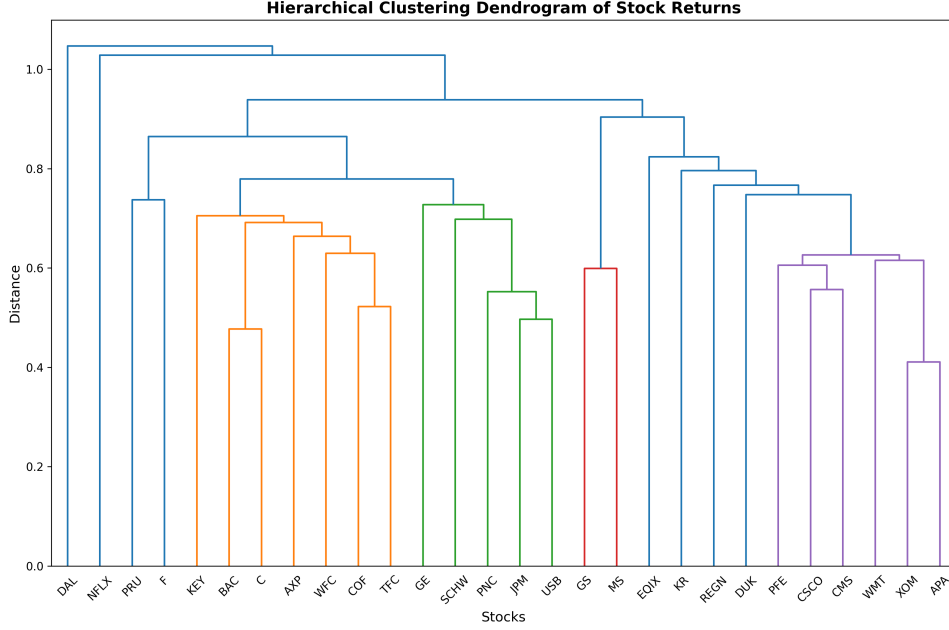


Figure 3: Dendrogram illustrating the hierarchical clustering of the 28 stocks based on correlation distances.

7 Performance Comparison of UCB1 and Epsilon-Greedy Algorithms

This section compares the performance of the Upper Confidence Bound (UCB1) and epsilon-greedy ($\epsilon = 0.1$) algorithms, implemented in Sections 5 and 6, using the daily log returns of 28 S&P 500 stocks over 42 trading days in September to October 2008. Performance is evaluated through cumulative wealth, regret (the difference between the maximum possible and actual returns), and variability (standard deviation of portfolio returns). Results are compared with those reported in [1] (Section 3.2, Figure 2), and key differences are summarised.

The UCB1 algorithm achieved a final cumulative wealth of 0.8126, outperforming the epsilon-greedy algorithm’s wealth of 0.7182 and the equal-weighted portfolio’s wealth of 0.7305. The UCB1 algorithm exhibited higher variability (0.0772) compared to epsilon-greedy (0.0690) and the equal-weighted portfolio (0.0547), indicating greater fluctuations in daily returns. The final cumulative regret was 4.1161 for UCB1, 4.2547 for epsilon-greedy, and 4.2756 for the equal-weighted portfolio, suggesting that UCB1 incurred the least regret by more effectively selecting high-return assets.

Figure 6 shows the cumulative wealth trajectories. The UCB1 portfolio generally maintained higher wealth, with notable growth in mid-October 2008, despite volatility during the financial crisis. The epsilon-greedy portfolio showed more erratic performance, likely due to random exploration ($\epsilon = 0.1$), while the equal-weighted portfolio remained stable but underperformed. Figure 7, illustrates cumulative regret, with UCB1 consistently showing lower regret, particularly after initial exploration, compared to the epsilon-greedy and equal-weighted portfolios.

In [1] (Section 3.2, Figure 2), the authors report that the UCB1 portfolio outperforms a risk-aware portfolio in stable markets but underperforms in volatile conditions, where the combined portfolio (balancing UCB1 and risk-aware strategies via λ) excels. Their

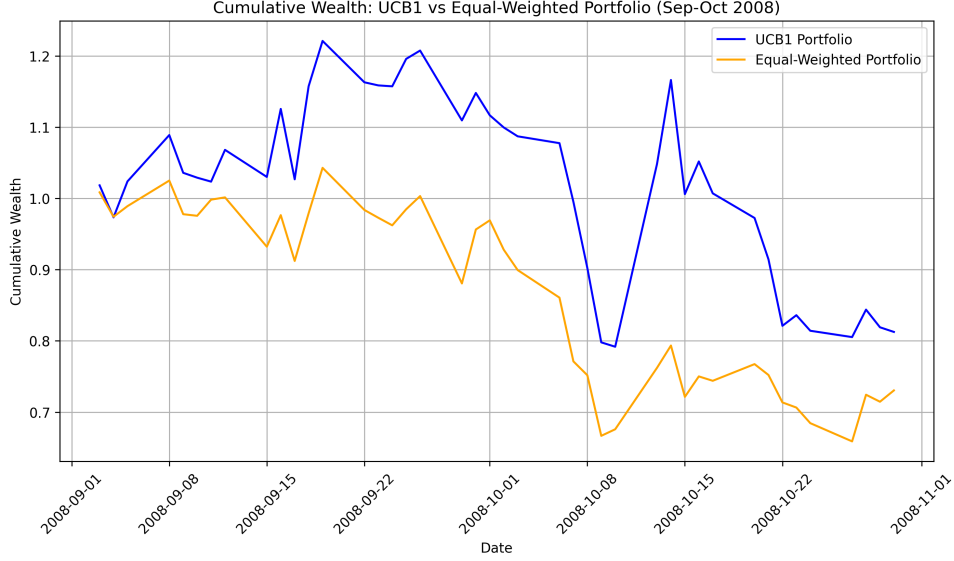


Figure 4: Cumulative wealth of the UCB1 portfolio compared to the equal-weighted portfolio over 42 trading days in September to October 2008.

simulations, using correlated geometric Brownian motions for 30 stocks, show the UCB1 portfolio achieving higher wealth than a benchmark in stable periods but with higher variance. In contrast, this study’s results show UCB1 outperforming both epsilon-greedy and equal-weighted portfolios in the volatile 2008 market, possibly due to the absence of the risk-aware component. The epsilon-greedy algorithm’s performance was less consistent, likely due to its random exploration, which occasionally selected suboptimal assets, as seen in the selection of APA and F in later trials.

7.1 Summary of Key Differences

The primary difference between this study and [1] lies in the market conditions and algorithm implementation. Huo and Fu simulate stock prices using a Black-Scholes model with controlled volatility, whereas this study uses real historical data from the 2008 financial crisis, a period of extreme volatility. Consequently, the UCB1 algorithm’s superior wealth (0.8126 versus 0.7182 for epsilon-greedy) contrasts with the paper’s findings, where the combined portfolio often outperformed UCB1 in volatile settings. The epsilon-greedy algorithm’s higher regret (4.2547) compared to UCB1 (4.1161) highlights its sensitivity to random selections, unlike the paper’s combined portfolio, which mitigates risk via CVaR minimisation. The equal-weighted portfolio’s low variability (0.0547) aligns with its role as a stable benchmark, consistent with the paper’s findings. Figures 6 and 7 visually confirm UCB1’s advantage in wealth accumulation and lower regret, driven by its adaptive selection strategy, despite the volatile market.

8 Data Collection and Processing for 2020

Daily closing prices for 28 of the 30 specified S&P 500 stocks were collected for March to April 2020. The stocks include 14 financial institutions (JPM, WFC, BAC, C, GS, USB, MS, KEY, PNC, COF, AXP, PRU, SCHW, TFC) and 14 non-financial companies (KR,

Algorithm 3 Epsilon-Greedy Algorithm for Portfolio Selection

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1: Input: Trials  $N$ , assets  $K$ , returns  $R_{i,t}$  for  $i = 1, \dots, K$ ,  $t = 1, \dots, N$ ,  $\epsilon$ 
2: Output: Portfolio weights  $\omega_t$ , wealth  $S_N$ 
3: Initialise  $n_i = 0$ ,  $\mu_i = 0$  for each asset  $i$ , wealth  $S_0 = 1$ 
4: for  $t = 1$  to  $N$  do
5:   Generate  $p \sim \text{Uniform}(0, 1)$ 
6:   if  $p < \epsilon$  then
7:     Select random asset  $i_t$ , set  $\omega_t = e_{i_t}$ 
8:   else
9:     Select  $i_t = \arg \max_i \mu_i$ , set  $\omega_t = e_{i_t}$ 
10:  end if
11:  Observe return  $R_{i_t,t}$ , update  $n_{i_t} \leftarrow n_{i_t} + 1$ 
12:  Update  $\mu_{i_t} \leftarrow \frac{(n_{i_t}-1)\mu_{i_t} + R_{i_t,t}}{n_{i_t}}$ 
13:  Update wealth  $S_t = S_{t-1} \cdot (1 + \omega_t^\top R_t)$ 
14: end for
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PFE, XOM, WMT, DAL, CSCO, EQIX, DUK, NFLX, GE, APA, F, REGN, CMS). Data for SIVB and PEAK were unavailable due to ticker changes or delisting. The prices were structured into a pandas DataFrame, and daily log returns were computed as $R_{i,t} = \log(P_{i,t+1}/P_{i,t})$.

The resulting DataFrame contains 41 trading days (3 March 2020 to 29 April 2020) and 28 stocks, capturing high volatility during the COVID-19 market crash. A time series plot of returns for JPM, BAC, NFLX, WMT, and GE shown in Figure 8, illustrates significant fluctuations, particularly in financial stocks (e.g., BAC with a minimum return of -0.167205) and energy stocks (e.g., APA with a minimum return of -0.773592).

9 UCB1 and Epsilon-Greedy Algorithms on 2020 Data

The Upper Confidence Bound (UCB1) and epsilon-greedy ($\epsilon = 0.1$) algorithms, as implemented in Sections 5 and 6, were applied to the daily log returns of 28 S&P 500 stocks over 41 trading days in March to April 2020, as described in Section 8. The algorithms selected portfolios to maximise cumulative wealth, starting with an initial wealth of 1, and their performance was compared against an equal-weighted portfolio.

The epsilon-greedy algorithm achieved a final cumulative wealth of 0.8354, outperforming the equal-weighted portfolio's wealth of 0.7897 and the UCB1 algorithm's wealth of 0.5177. The UCB1 algorithm initially explored diverse stocks (e.g., JPM, WFC, BAC) before selecting assets such as NFLX and WMT, while the epsilon-greedy algorithm frequently selected MS in later trials, reflecting its exploitation of high-return assets balanced with random exploration. A plot comparing the cumulative wealth of the UCB1, epsilon-greedy, and equal-weighted portfolios, shown in Figure 9. The plot illustrates that the epsilon-greedy portfolio exhibited stronger growth, particularly in April 2020, while the UCB1 portfolio experienced significant declines, likely due to poor asset selections during the volatile COVID-19 market crash.

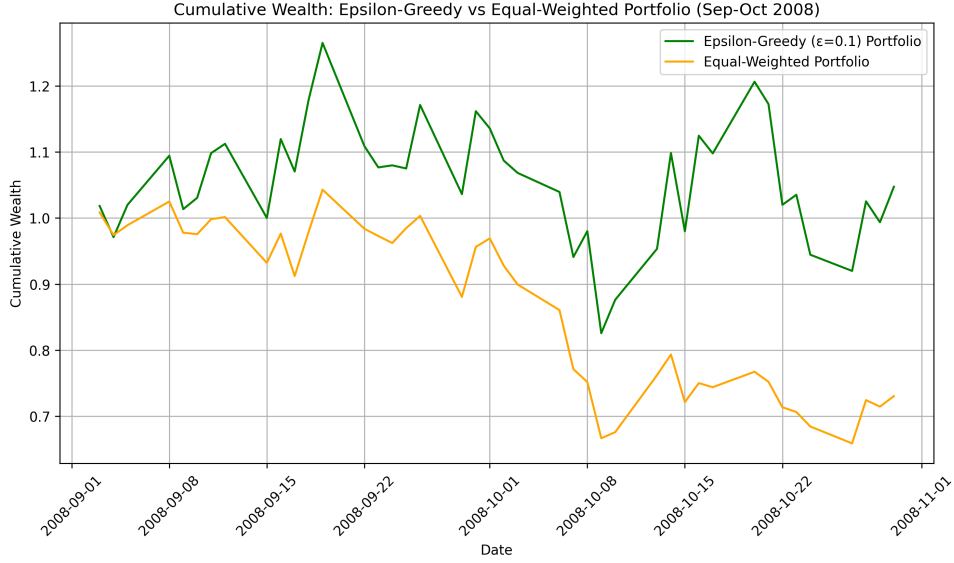


Figure 5: Cumulative wealth of the epsilon-greedy portfolio ($\epsilon = 0.1$) compared to the equal-weighted portfolio over 42 trading days in September to October 2008.

References

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- [2] Rosario N. Mantegna. Hierarchical structure in financial markets. *The European Physical Journal B*, 11(1):193–197, 1999.
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- [4] F. Pozzi, T. Di Matteo, and T. Aste. Spread of risk across financial markets: Better to invest in the peripheries. *Scientific Reports*, 3:1665, 2013.

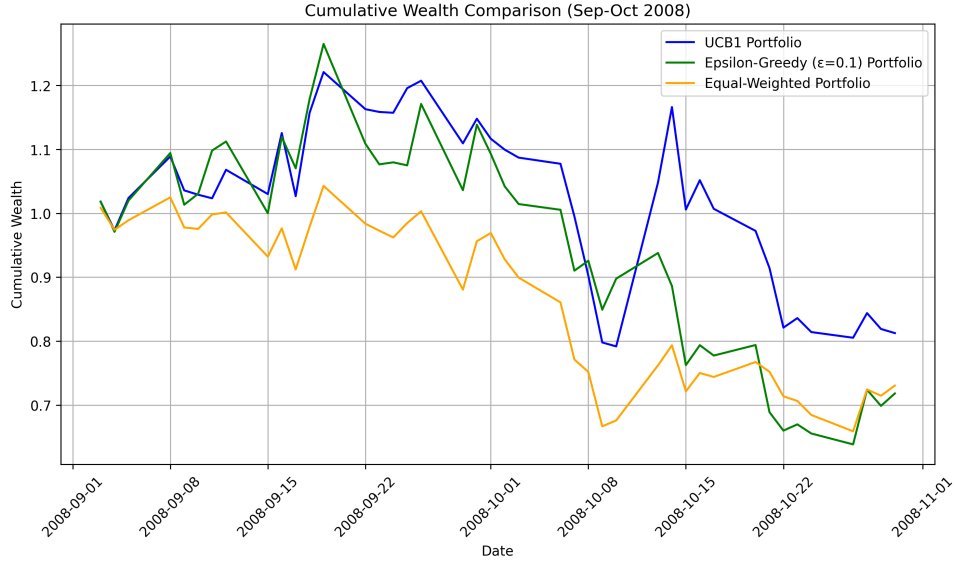


Figure 6: Cumulative wealth of UCB1, epsilon-greedy ($\epsilon = 0.1$), and equal-weighted portfolios over 42 trading days in September to October 2008.

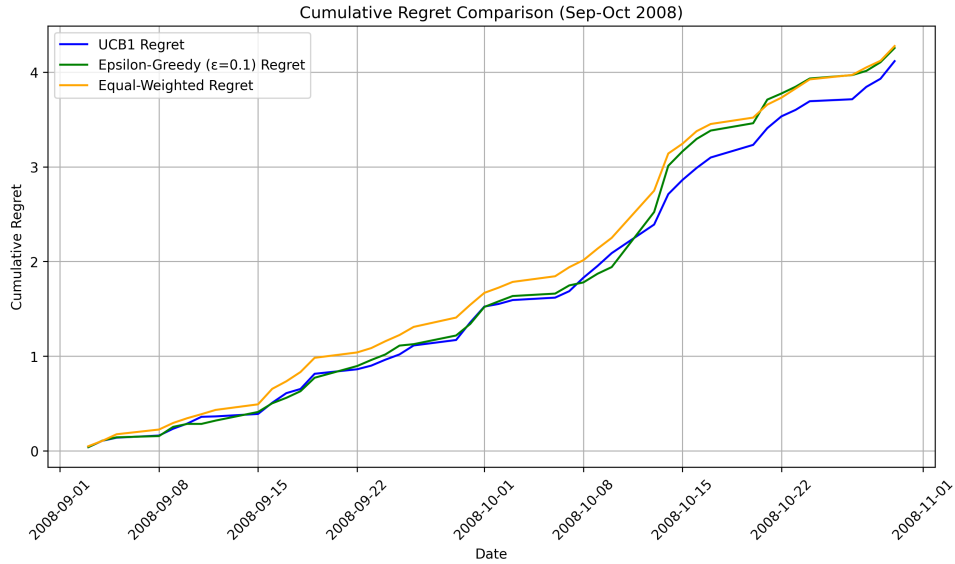


Figure 7: Cumulative regret of UCB1, epsilon-greedy ($\epsilon = 0.1$), and equal-weighted portfolios over 42 trading days in September to October 2008.

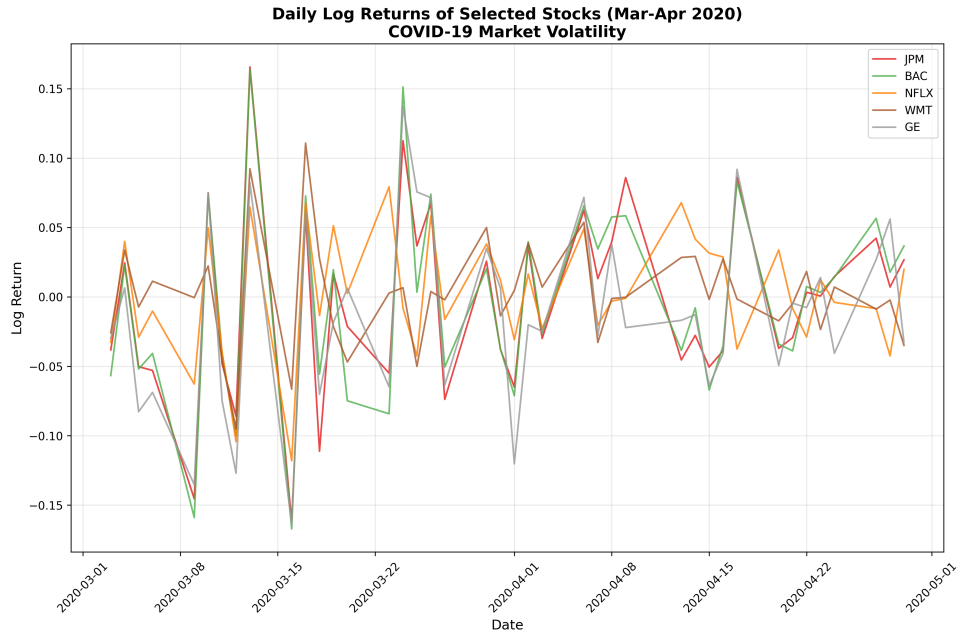


Figure 8: Daily log returns of selected stocks (JPM, BAC, NFLX, WMT, GE) over 41 trading days in March to April 2020.

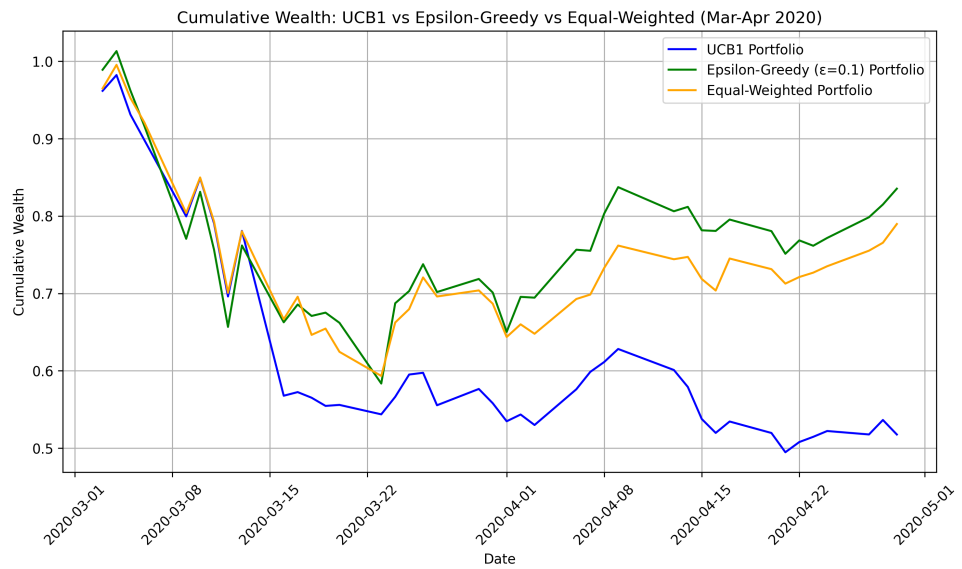


Figure 9: Cumulative wealth of UCB1, epsilon-greedy ($\epsilon = 0.1$), and equal-weighted portfolios over 41 trading days in March to April 2020.